Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)		Use of Doppler equation (1) Obtaining data from graph i.e. mean shift = $(4.6 \pm 0.1) \times 10^{-15}$ m **OR** max and min = 5.55 [$\times 10^{-15}$ m] and 3.65 [$\times 10^{-15}$ m] **OR** max = 5.55 [$\times 10^{-15}$ m] and 2.54 [m s^{-1}] (1) Amplitude = $(0.95 \pm 0.1) \times 10^{-15}$ [m] **OR** velocities = 1.67 and 2.54 [m s^{-1}] **OR** minimum = 3.65 [$\times 10^{-15}$ m] and 1.67 [m s^{-1}] (1) All penultimate steps seen or correct values to extra d.p. seen e.g. $\frac{3 \times 10^{8} \times 0.95 \times 10^{-15}}{656 \times 10^{-9}}$ **and** $\frac{3 \times 10^{8} \times 4.6 \times 10^{-15}}{656 \times 10^{-9}}$ (1) N.B. obtaining 0.43 [m s^{-1}] **or** 2.1 [m s^{-1}] gains the 1st 3 marks	1	1 1 1		4	3	
	(b)		Blue **and** shorter wavelength (or squashed or equivalent or negative shift) Accept negative velocity Don't accept because it is negative	1			1		
	(c)		Use of $v = \frac{2\pi R}{T}$ **OR** equivalent ($v = \omega R$ **and** $\omega = \frac{2\pi}{T}$) (1) T = 390 ± 10 days obtained from graph (1) Final sub seen **or** correct extra dp seen e.g. $R = \frac{0.4 \times 390 \textbf{ ecf} \times 24 \times 3600}{2\pi}$ **OR** 2145 km etc (1) N.B. use of this equation is incorrect $d^3 = \frac{GMT^2}{4\pi^2}$		3		3	2	